RRC Management System: A Comprehensive Web-based Solution for Workforce Deployment, Sales Monitoring, Inventory Management, and Client Services

A Capstone Project
Presented to the Faculty of the
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In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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ENDORSEMENT FORM FOR ORAL DEFENSE

TITLE OF RESEARCH: RRC Management System: A Comprehensive Web-based Solution for Workforce Deployment, Sales Monitoring, Inventory Management, and Client Services

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This capstone project titled: RRC Management System: A Comprehensive Web-based Solution for Workforce Deployment, Sales Monitoring, Inventory Management, and Client Services prepared and submitted by **Dan Adrian A. Austria, Edgar Joseph M. Basilan, Dan Fredrick E. Dela Cruz**, and **Zeth Nikolai J. Pagarigan**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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INTRODUCTION

Project Context

The pest control industry is vital for maintaining public health, sanitation, and structural integrity in both residential and commercial settings. However, many pest control companies continue to rely on outdated, manual processes for workforce deployment, inventory management, and client transactions. These traditional methods—such as paper-based record-keeping and phone-based scheduling—are prone to inefficiencies, miscommunication, and operational delays, leading to decreased service reliability and customer dissatisfaction. The pest control market is projected to reach $36.1 billion by 2030, largely due to advancements in digital tools and automation.

With the increasing demand for precision, speed, and accountability in pest control services, adopting a web-based management system has become essential. The RRC Management System is designed to address these industry challenges by integrating workforce tracking, automated scheduling suggestions, secure online payments, and real-time transaction management, ensuring a seamless experience for both administrators and existing clients. The system leverages modern technologies such as ASP.NET for backend processing, MS SQL for database management, and advanced cybersecurity protocols to maintain data integrity. To further enhance security, the system implements Argon2 password hashing, AES-256 encryption, blockchain technology for transaction transparency, and automated database backups to protect sensitive company and client data from cyber threats.

By automating key business processes, the proposed system aims to enhance productivity, minimize operational inefficiencies, and improve client satisfaction. The structured digital approach of the RRC Termite & Pest Control Management System aligns with best practices in digital service management and contributes to industry-wide improvements in workforce efficiency and business transparency. This transition to a centralized, technology-driven framework not only modernizes pest control operations but also serves as a model for leveraging digital innovation to optimize service-based industries.

Purpose and Description of the Project

The RRC Management System is a comprehensive, web-based solution designed to modernize and centralize the management of pest control services. Traditional pest control businesses often rely on manual processes, such as paper-based record-keeping, phone-based scheduling, and decentralized workforce deployment. These outdated methods lead to inefficiencies, operational delays, miscommunication, and difficulty in tracking inventory and financial transactions.

To address these challenges, the RRC Management System introduces an integrated digital approach that automates essential business processes, minimizes human errors, and enhances operational efficiency. The system enables seamless transactions, workforce tracking, inventory monitoring, optimized service scheduling suggestions, and secure online payment processing through PayPal, ensuring a well-coordinated and transparent business operation accessible to verified existing clients.

By leveraging modern technologies, such as ASP.NET for backend processing, MS SQL for secure database management, and advanced cybersecurity measures (including Argon2 password hashing, AES-256 encryption, and blockchain technology), the system ensures high data integrity and protection. Workforce tracking features allow dispatchers to effectively deploy pest control teams within designated areas, reducing downtime and improving service efficiency. Clients benefit from a streamlined service experience, including the ability to schedule free inspections, send image-based inquiries, and settle payments online with ease.

Furthermore, the system enhances administrative oversight through tiered security access levels, where the System Administrator can regulate permissions and monitor activity logs. The Business Administrator efficiently manages transactions and inventory, while the Dispatcher ensures smooth workforce operations. Maintenance personnel handle product and service updates, reinforcing adaptability in a fast-changing industry. Meanwhile, clients experience improved service reliability, fostering trust and long-term customer engagement.

Ultimately, the RRC Management System aims to enhance productivity, streamline operational workflows, and increase transparency in both financial and service-related transactions. By integrating automation, secure online payments, and digital inspection assignment tools, the system modernizes pest control operations, setting a new industry benchmark for efficiency, security, and customer satisfaction.

Objectives of the Study

- To develop a web-based management system for RRC Termite & Pest Control to enhance record-keeping, workforce deployment, sales tracking, inventory management, and scheduling processes**.**
- To enhance workforce deployment by providing a system that assists in tracking and suggesting inspector assignments based on service location.
- To develop a user-friendly web-based solution using ASP.NET for backend processing, MS SQL for database management, Blockchain technology, Argon2 password hashing, AES-256 encryption, and database backups for the cybersecurity measures**.**
- To improve inventory management by tracking the availability and usage of equipment and chemicals.
- To optimize client interaction by enabling existing clients to access transaction history, check balances, schedule services, send proof of inquiries with images, and securely settle payments through PayPal.
- To increase sales transparency by offering a structured view of transaction history and analytics for enhanced financial management.
- To centralize administrative control by enabling the System Admin to manually manage accounts, assign permissions, and regulate user access levels.
- To implement an inspection-first workflow where inspectors input service requirements and generate quotations based on actual site assessments.

Scope and Limitations of the Study

Scope of the Study

The system includes the following features:

- **Inspection-Based Workflow:** The system facilitates service workflows starting from client inquiries to inspection scheduling and quotation generation. Clients can submit inquiries and schedule a free inspection. Only after inspection will service quotations be provided and further steps be taken.
- **Client Interactions:**
    - Submit service inquiries
    - Schedule inspection appointments
    - Upload supporting images for inspection
    - Review the company's Terms of Service
    - Make online payments for accepted quotations via PayPal
    - Clients cannot create their own accounts, directly schedule pest control services, edit service details, or communicate with contractors within the system. The Business Admin is responsible for client account creation and clients are only permitted to submit booking requests after inspection approval.
- **Inspection Scheduling and Assignment Suggestion:** Inspectors are manually assigned by dispatchers. The system only suggests suitable inspectors based on proximity and availability but does not automatically assign them. Daily inspections are capped at a maximum of 6 for each inspector. Failed inspections (e.g., unclear site scope) are flagged for review and accounted for by emergency reserves and buffers for each operations.
- **Administrative and System Control:** The System Administrator can manually create user accounts, manage user permissions, and control access across all system modules. Role-based access is dynamically configured, and some modules may be hidden based on user role permissions. All user login

credentials are securely hashed using Argon2 and are inaccessible for review. Passwords are set by the user after account creation through e-mail.

- **Inventory Management:** Admins and dispatchers can monitor chemical and equipment stock levels, usage history, and add new stock for pest control operations. Emergency reserves or buffers can also be accounted for in anticipation of unexpected demand.
- **Audit Trail and Inquiry Logs:** All system activities, especially those involving sensitive data or administrative actions, are recorded in audit logs for security, compliance, and administrative oversight. Previous records are not visible to users after submission for added data privacy.
- **Reports and Analytics:** The system provides data summaries and operational insights, including workforce activity, quotation records, and service outcomes. These help administrators assess performance and support business planning
- **Backup and Recovery:** Secure and automated database backups are implemented to support disaster recovery. Critical data including inspection records, client transactions, and inventory logs are preserved to ensure business continuity.

Limitations of the Study

While the RRC Termite & Pest Control Management System introduces significant improvements in operational efficiency, scheduling, and security, the following limitations were intentionally implemented to align with the current capabilities and scope of the company:

- **Geographical Limitations:** The system only suggests deployment within adjacent areas to the company's main branches, and no automated assignment is done.
- **Administrator-managed Client Creation:** Only system administrators have the capability to create and manage client accounts; self-registration is not supported.
- **Limited Online Payment Functionality:** The online payment functionality is limited to PayPal; other payment gateways are not currently integrated.
- **No Live Inspector Auto-Assignment:** Inspection modules require manual assignment, as the system only offers deployment suggestions.
- **Security Scope:** 2FA only applies to System Admin and other roles.
- **Client Booking Restriction:** Clients are limited to booking and inquiry submission; inspectors input service details post-inspection.

By implementing this innovative and secure digital solution, the RRC Termite & Pest Control Management System aims to set a new benchmark for efficiency and transparency in the pest control industry.

REVIEW OF RELATED LITERATURE/SYSTEMS

Review of Related Literature

Global Studies

Cybersecurity Risk Management

As companies digitize their business operations, cyber risks increase exponentially. A study by McKinsey & Company (2023) highlights that data breaches, malware attacks, and insider threats pose significant risks to service-based industries, including pest control. Effective cybersecurity strategies, such as encryption, multi-factor authentication, and continuous monitoring, are crucial for safeguarding client and business data. However, cybersecurity solutions can introduce complexities in system management and maintenance, requiring businesses to allocate additional resources to IT security.

Cybersecurity in the Digital Era

The increasing reliance on cloud computing and interconnected digital platforms has made cybersecurity a top priority in service industries. Cyberattacks not only lead to financial loss but also damage a company's reputation and customer trust (PestSure, 2023). Implementing blockchain technology and AI-driven security measures can significantly mitigate these threats. While blockchain ensures tamper-proof transaction logs, its adoption in pest control is still in its early stages, requiring further research on feasibility and scalability.

Argon2 Memory-Hard Function for Password Hashing and Proof-of-Work Applications

Argon2 is a memory-hard function. It is a streamlined design. It aims at the highest memory-filling rate and effective use of multiple computing units, while still providing defense against trade-off attacks. Argon2 is optimized for the x86 architecture and exploits the cache and memory organization of the recent Intel and AMD processors. Argon2 has one primary variant, Argon2id, and two supplementary variants, Argon2d and Argon2i. Argon2d uses data-dependent memory access, which makes it suitable for cryptocurrencies and proof-of-work applications with no threats from side-channel timing attacks. Argon2i uses data-independent memory access, which is preferred for password hashing and password-based key derivation. Argon2id works as Argon2i for the first half of the first pass over the memory and as Argon2d for the rest, thus providing both side-channel attack protection and brute-force cost savings due to time-memory trade-offs. Argon2i makes more passes over the memory to protect from trade-off attacks (Biryukov, et al., 2022).

Development of a Web-Based Platform for Automating an Inventory Management of a Small and Medium Enterprise

Nowadays, it has been noticed that the majority of businesses manually record and maintain their inventory data in spreadsheets with little to no technological innovation. This has many drawbacks in our contemporary society because it has been demonstrated that successfully monitoring inventories is difficult. This manual method of record keeping was labor-intensive, expensive, and error-prone, and it was unable to guarantee that the inventory remained current due to oversight and internal shrinkage. This study investigates the problems with manual inventory management and creates computerised inventory management software to solve those issues. By creating a computerised inventory management system to help storekeepers make decisions about their stocks, it suggests solutions to the current problems by keeping the records, tracking employee salaries, and updating sales and transactions (Agboola, et, al., 2022).

AES-256 Encryption for Secure Data Protection

AES-256 (Advanced Encryption Standard with a 256-bit key) is a symmetric encryption algorithm widely recognized for its high level of security and performance. It encrypts data in fixed 128-bit blocks and utilizes a key schedule to derive round keys from the original key. AES-256 provides 14 rounds of substitution and permutation operations, making brute-force attacks computationally infeasible with current technology. Due to its strength and efficiency, AES-256 is often used in government, financial, and healthcare applications to secure sensitive information. It is resilient against most known attacks, including differential and linear cryptanalysis, and is recommended by the U.S. National Institute of Standards and Technology (NIST) for securing classified information (Wang et al., 2024).

Two-Factor Authentication for Account Security

Two-Factor Authentication (2FA) is a layered security mechanism that requires users to provide two separate forms of identity verification before gaining access to a system. Typically, 2FA combines something the user knows (e.g., a password) with something the user has (e.g., a mobile phone or hardware token). This dual-layer approach significantly reduces the risk of unauthorized access caused by stolen or guessed passwords. Common implementations of 2FA include time-based one-time passwords (TOTP) delivered via mobile apps or SMS. The technology is widely adopted across banking, email, and enterprise systems to enhance user authentication protocols (Tetlay et al., 2020).

PayPal Integration for Online Payment Systems

PayPal is a widely-used online payment gateway that facilitates secure financial transactions over the internet. It allows users to send and receive money without directly sharing sensitive financial details such as bank or credit card numbers. PayPal supports multiple currencies and payment methods and is equipped with fraud detection systems, encryption protocols, and buyer/seller protection mechanisms. Developers can integrate PayPal into web applications through APIs, providing features such as checkout processing, payment confirmations, and automated refunds. Due to its global reach and ease of use, PayPal is a preferred choice for e-commerce, subscription services, and SaaS platforms (The Wall Street Journal, 2025).

Microsoft SQL Server for Relational Database Management

Microsoft SQL Server (MS SQL) is a relational database management system (RDBMS) developed by Microsoft that supports a wide range of data processing and transaction capabilities. It is built to handle large volumes of structured data, enabling efficient storage, retrieval, and management of records through SQL (Structured Query Language). MS SQL supports advanced features such as indexing, stored procedures, triggers, views, and real-time transaction processing. It also includes robust tools for backup, recovery, and security management. The integration with .NET frameworks makes it ideal for enterprise-level applications requiring high availability, consistency, and scalability (Talekattu et al., 2024).

Role-Based Access Control (RBAC) for Secure Permission Management

Role-Based Access Control (RBAC) is a security model that restricts system access based on a user's role within an organization. Instead of assigning permissions to individual users, access rights are grouped by role, and users are assigned to roles according to their responsibilities. This model simplifies permission management, enhances security by enforcing the principle of least privilege, and supports regulatory compliance by providing structured access control. RBAC is particularly effective in complex systems where many users interact with sensitive data and varied functionality. It is widely implemented in enterprise systems, cloud services, and healthcare applications due to its scalability and auditability (Hu et al., 2023).

ASP.NET for Web Application Development

ASP.NET is an open-source, server-side web application framework developed by Microsoft, designed for building dynamic and interactive web applications. It supports multiple programming models including MVC (Model-View-Controller) and Web API for scalable and modular development. ASP.NET applications run on the .NET runtime and can integrate seamlessly with databases, authentication systems, and third-party services. With features like session state, routing, model binding, and built-in security (e.g., role-based access), ASP.NET simplifies the creation of secure and maintainable web solutions. Its compatibility with Visual Studio enhances productivity through IntelliSense, debugging, and deployment tools (Microsoft Developer Division, 2023).

Local Studies (Philippines)

AES-256 Encryption in Philippine SMEs

AES-256 encryption plays a critical role in protecting sensitive data within businesses in the Philippines. ITx.PH emphasized that AES-256 helps local SMEs secure personal and financial data, ensuring compliance with the Data Privacy Act of 2012. The site stresses that strong encryption methods like AES-256 should be standard in local digital platforms that handle customer information and payment records (ITx.PH, n.d.).

The Influence of Inventory Management on Operational Performance of Manufacturing Businesses in Tagum City

Inventory management is a crucial component in ensuring operational success, especially in manufacturing businesses. This study examined the correlation between inventory practices and operational performance in Tagum City. Findings revealed that sound inventory strategies, including proper stock monitoring and order scheduling, directly contributed to improved efficiency and reduced operational issues in the manufacturing sector. (Gaid, Pagcas, & Pagdilao, 2022)

Microsoft SQL Server in Web-Based Inventory Systems

In a study on digital transformation in inventory processes, researchers from Pagadian City implemented a web-based inventory management system utilizing Microsoft SQL Server for efficient data handling. The system enabled accurate, real-time inventory tracking and minimized human error, making it a viable solution for SMEs in the region. SQL Server was chosen due to its scalability and security features, aligning with local needs for reliable business tools (Gumapon & Cabanlit, 2023).

Cybersecurity Challenges in the Philippines

According to OpenGov Asia, the Philippines suffers from a lack of cybersecurity professionals, with only around 200 certified experts nationwide. This shortage poses risks to businesses transitioning to online systems. As local enterprises continue digitalization, the use of encrypted systems, secure access protocols, and awareness of evolving threats becomes increasingly essential (OpenGov Asia, 2022).

Role-Based Access Control (RBAC) Practices

While formal studies on RBAC in the Philippines are limited, its practical application in access management systems is noted in business software design. RBAC ensures that users can only access data and functions necessary for their role, thereby reducing potential abuse or data breaches. This control mechanism is often implemented alongside web technologies used in inventory systems and financial applications to uphold internal security (ITx.PH, n.d.).

Two-Factor Authentication (2FA) Adoption in the Philippines

A regional study by Kaspersky revealed that 75% of Filipino digital banking users prefer one-time passwords (OTPs) via SMS for transaction verification, highlighting a strong inclination towards 2FA mechanisms to safeguard financial transactions (Manila Bulletin, 2022). Furthermore, a survey conducted by GSMA indicated that 71.4% of Filipinos believe that account security risks are increasing, with financial fraud being among their top concerns. The study emphasized the need for enhanced security measures, including 2FA and SIM-swap prevention, to protect consumers from various forms of fraud (BusinessWorld Online, 2024).

PayPal Integration in Philippine Social Commerce

According to a study highlighted by Adobo Magazine (2025), 87% of Filipino merchants engaged in social commerce accept digital payments, with PayPal being the most used platform. The integration of PayPal has facilitated secure and efficient transactions, contributing to improved financial conditions for both merchants and consumers in the Philippines.

A System Study on the Inventory Management and Warehousing Operations of Oriental Merchants Inc.

This study investigated the inventory and warehousing operations of Oriental Merchants Inc., identifying bottlenecks and inefficiencies in product handling. It proposed a revised layout and scheduling system to enhance inventory flow, reduce returns, and ensure timely deliveries to clients. The results supported the integration of process improvements in inventory systems. (Reyes, Tomacruz, & Yang, 2016)

ASP.NET in Local Web Development

ASP.NET has gained traction among Filipino developers building robust web-based systems, including inventory and financial management platforms. Its support for secure authentication methods, integration with SQL Server, and scalability makes it suitable for government and private-sector applications. Several academic capstone projects and systems in local universities use ASP.NET due to its compatibility with enterprise-grade technologies (Gumapon & Cabanlit, 2023).

Argon2 Password Hashing for Enhanced Security

Argon2, the winner of the 2015 Password Hashing Competition, is recognized for its resistance to brute-force attacks and its memory-hard design, making it suitable for securing user credentials. While specific studies in the Philippine context are limited, the adoption of Argon2 in various applications underscores its effectiveness in enhancing password security (Davro.net, 2023).

Related Studies and/or Systems

Global Systems

Libsodium

Argon2id is implemented by a cross-platform, memory-hard password hashing API with safe defaults (1 pass, 4 MiB memory, 3 iterations). It makes it easier to store user credentials securely in any language with a C binding by automatically creating a 32-byte hash and a 16-byte salt. All that is needed for integration is to include and link the single libsodium library.

HashiCorp Vault Transit Secret Engine

Argon2ID hashing, AES-GCM encryption, and HMAC are among the managed, HSM-backed cryptographic operations that Vault's Transit engine offers via a REST API. It also enforces policy-driven access controls, rotates keys automatically, and logs audits, making it perfect for centralized secret management in multi-tenant environments.

AWS Key Management Service (KMS)

A fully managed service that uses AES-256-GCM to perform envelope encryption with AWS-managed or customer-managed CMKs. It integrates with S3, RDS, and Lambda for transparent at-rest encryption and manages key lifecycle, rotation, and access control through IAM policies.

Hyperledger Fabric

A permissioned, modular DLT framework that supports fine-grained access control through MSPs and pluggable consensus (Solo, Raft, and IBFT). Chaincode (smart contracts) are ideal for securely tracking transactions in business supply chains because they can generate immutable audit logs while operating in Docker containers.

R3 Corda

Pluggable consensus, notaries for double-spend protection, and ID mapping are all enforced by this DLT platform, which is perfect for financial transaction workflows that require privacy and compliance. It is made for regulated environments; there is no global broadcast, only point-to-point states via "flows."

Oracle NetSuite Inventory Management

Oracle NetSuite offers a comprehensive cloud-based inventory management solution that provides real-time visibility into inventory levels, order management, and supply chain operations. This system enables businesses to optimize stock levels, reduce carrying costs, and improve overall operational efficiency. Its scalability makes it suitable for businesses of various sizes, adapting to complex inventory needs as companies grow (Oracle NetSuite, n.d.).

SAP Extended Warehouse Management (SAP EWM)

SAP EWM is designed for complex warehouse operations, offering advanced features like automated picking, slotting, and real-time inventory tracking. It enhances warehouse efficiency by integrating with other SAP modules, facilitating seamless supply chain management. The system supports both inbound and outbound processes, ensuring accurate inventory management and timely order fulfillment (SAP, n.d.).

Walmart's Retail Link System

Walmart's Retail Link is a proprietary system that allows suppliers to access real-time sales and inventory data, facilitating efficient restocking and supply chain collaboration. By providing transparency and data-driven insights, it enables suppliers to make informed decisions, aligning production and distribution with consumer demand (Walmart, n.d.).

Workday

Workday is a unified platform offering human capital management, financial management, and planning solutions, enhanced by AI-driven insights to optimize workforce planning and deployment. Its recent introduction of Illuminate AI Agents and the Agent System of Record (ASR) aims to automate routine tasks, improve decision-making, and maintain human oversight in critical processes (TechRadar, 2025).

GoCo

GoCo is an all-in-one HR solution tailored for small to mid-sized businesses, streamlining processes like onboarding, benefits administration, and time tracking. Its customizable workflows and integrations with other HR tools make it a flexible solution for managing workforce operations efficiently (GoCo, n.d.).

Local Systems (Philippines)

ZeroPest PH's Digital Pest Control

ZeroPest PH integrates advanced monitoring systems and environmentally safe pest management solutions. Their approach combines preventive measures, real-time tracking, and targeted interventions to ensure effective pest control with minimal environmental impact (ZeroPest PH, n.d.). However, challenges in scaling operations and ensuring cost efficiency remain key concerns.

Ensystex Philippines' Urban Pest Management System

Ensystex focuses on providing environmentally responsible digital pest management solutions for urban areas. Their system combines remote monitoring and automated reporting, ensuring professional pest managers can act swiftly when necessary (Ensystex PH, n.d.).

Environet Pest Control Website

Environet Pest Control provides pest control services in the Philippines, they are a termite control company established in 2000. Their system allows for getting a free quote for their services, along with a free inspection (Environet Pest Control, n.d.).

Web-based Inventory Management System

This software project is a web-based inventory management system for a small business enterprise established in Pagadian City, Zamboanga del Sur, Philippines. It primarily supports the business owner of the said business enterprise to perform regular inventory management online through a web platform. This mechanism did not only improve the overall business process but also enabled the business owner to manage inventories with efficiency and convenience. The enterprise comprises four branches and a mobile store that strolls around the city to cater to customer needs. In addition, the current mechanism of conducting inventory management is purely manual through the use of paper files, which overwhelmed the business owner with several challenges, including data inaccuracy due to the reliance on using papers where data are recorded and process inefficiency due to the manual recording and distribution of inventory and sales reports. This software project addressed these challenges by eliminating these manual approaches through an automated computing solution using a web platform where data are electronically recorded and processed, and reports are electronically produced (Tanaman, et al., 2023).

Due to the limited availability of recent and reputable local systems, the proponents have included older systems to further substantiate the validity and foundation of this project.

Synthesis

The reviewed literature reveals a consistent shift among Philippine organizations toward secure and efficient web-based systems. AES-256 encryption emerges as a cornerstone for data protection, especially in platforms handling sensitive customer and financial data. Its adoption is critical in meeting the compliance requirements of the Data Privacy Act of 2012. Complementing this is Microsoft SQL Server, which provides scalable and secure data storage, making it an effective choice for systems like web-based inventory management, as demonstrated by academic projects in local institutions.

Cybersecurity remains a pressing concern in the Philippines due to the low number of certified experts, highlighting the need for stronger built-in security mechanisms in systems developed locally. This is reflected in the use of role-based access control (RBAC) for internal security, ensuring that only authorized users can access specific system functionalities based on their roles.

The widespread use of two-factor authentication (2FA) in digital banking—especially through one-time passwords (OTPs) sent via SMS—indicates a high level of local user acceptance and need for multi-layered security. Additionally, Argon2 password hashing is gaining recognition for its robustness against brute-force attacks and suitability for safeguarding user credentials, providing an added layer of protection for authentication systems.

In the field of digital transactions, PayPal integration has proven essential for Filipino merchants, particularly in the context of social commerce. Studies show that PayPal not only supports secure online payments but also contributes to improving the financial health of users, making it an ideal payment gateway for local applications.

Lastly, ASP.NET is highlighted for its flexibility and enterprise-grade features, supporting both security protocols and seamless integration with other Microsoft technologies. Its local adoption in academic and government system development showcases its relevance and capability in delivering secure, scalable web solutions.

Together, these technologies and practices reflect an ecosystem moving toward more secure, accessible, and functional web-based platforms in the Philippines, especially for applications like inventory and financial management systems.

TECHNICAL BACKGROUND

Overview of Current Technologies to be Used in the System

The RRC Pest Control Management System is built using a combination of modern web technologies, secure data handling methods, and scalable infrastructure to ensure operational efficiency and system integrity. It is developed as a web-based application, integrating both frontend and backend technologies to deliver a seamless user experience and effective data management.

The backend is developed using ASP.NET, which facilitates the implementation of complex business logic and robust API development. This framework ensures secure and efficient communication between the frontend interface and the underlying database. The system uses Microsoft SQL Server (MS SQL) as its database management system, offering a structured and high-performance platform for data storage. MS SQL supports stored procedures, indexing, and real-time transaction processing, which are essential for handling operational data with speed and reliability.

To maintain high standards of data protection, the system incorporates multiple security protocols. Argon2 is employed for secure password hashing, reducing the risk of credential theft. Additionally, AES-256 encryption is applied to protect sensitive data such as client information, uploaded files, and digital receipts, ensuring data confidentiality and integrity. Two-factor authentication (2FA) further strengthens account security by requiring a secondary form of verification during login.

Role-Based Access Control (RBAC) is enforced throughout the system to define and limit user permissions based on roles, thereby minimizing the risk of unauthorized access and supporting operational accountability. The platform is deployed using cloud-based hosting, which enables remote accessibility, scalability, and system availability under varying workloads. Automated backups and disaster recovery mechanisms are integrated to prevent data loss and ensure continuity of operations in the event of technical failure.

To support modern financial transactions, the system integrates PayPal, a secure online payment gateway, enabling clients to make real-time payments through various channels such as credit cards, e-wallets, and online banking. This integration streamlines billing processes and enhances the convenience and flexibility of service payments.

Collectively, these technologies provide a secure, scalable, and reliable infrastructure that supports the system's goals of improving pest control workflows, optimizing resource management, and enhancing overall client service delivery.

Calendar of Activities

The calendar of activities outlines the key milestones and tasks completed throughout the development of the RRC Management System. It details the sequence of activities, their objectives, the individuals involved, and the necessary resources used to ensure the successful completion of the project. This structured timeline provides an overview of the planning, research, development, and evaluation phases undertaken by the proponents.

February 2-8: The proponents brainstormed potential titles for the proposal. Initial research was conducted on various topics to ensure alignment with current industry trends and feasibility. Resources used included academic journals, online databases, and consultations with faculty members. Persons involved: all team members.

February 9-11: The initial creation of Chapter 1 was completed based on the selected titles. This stage involved drafting the introduction, problem statement, objectives, and significance of the study. Resources used included thesis writing guidelines, previous research studies, and adviser recommendations. Persons involved: all team members.

February 12: The first meeting with the thesis adviser, was conducted to discuss the proposal. The goal was to receive initial feedback and ensure the research was aligned with academic requirements. Resources used: prepared draft of Chapter 1 and research materials. Persons involved: all team members and the thesis adviser.

February 13-14: Revisions of Chapter 1 were made based on feedback from the first meeting. The proponents refined the content, addressing gaps or inconsistencies highlighted by the adviser. Resources used: thesis guidelines, adviser comments, and research articles. Persons involved: all team members.

February 15: The title defense was conducted. The proponents presented their proposed study before a panel, defending the relevance, feasibility, and significance of the research. The objective was to gain approval for the study. Resources used: presentation slides, research documents, and visual aids. Persons involved: all team members, thesis adviser, and panel members.

February 16-25: Exam Week; the development of the prototype for the database and system was initiated, along with continued research for the chosen title. The objective was to begin the practical implementation of the project while simultaneously gathering relevant literature to support the study. Resources used: database management software, programming tools, research papers, and consultation with IT experts. Persons involved: all team members.

February 26: The first official consultation with the adviser was conducted, and the initial database and system development progress was reviewed. The adviser provided feedback for improvements. Resources used: system prototype, database schema, documentation, and consultation notes. Persons involved: all team members and the thesis adviser.

March 5: The second official consultation with the adviser was conducted. The improvements made in the database and system were assessed, and the literature review section (Chapter 2) was reviewed for coherence and relevance. Resources used: updated system prototype, revised Chapter 2, adviser comments, and academic references. Persons involved: all team members and the thesis adviser.

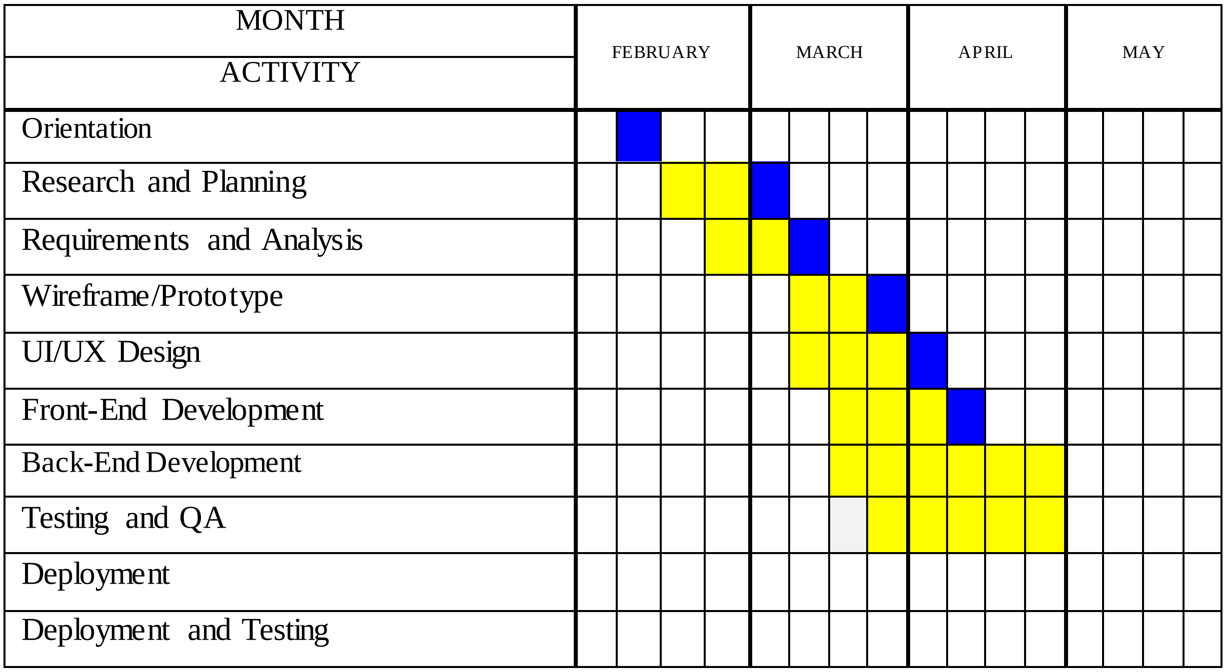
March 12: The third official consultation with the adviser was conducted, and the focus was to further enhance the system and to also prepare for the Chapter 3 of the paper, drafting the necessary technological background graphs needed. Resources used: updated system prototype, Chapter 3 creation, adviser comments and flowchart, ERD. Persons involved: all team members and thesis adviser.

March 19: The fourth official consultation with the adviser was conducted. There were minor adjustments needed to include the necessary details regarding the equipments and to the Login Module, as well as establishing clearer understanding towards the Process and Client Experience. Resources used: updated system prototype, adviser comments. Persons involved: all team members and thesis adviser.

April 5: The fifth official consultation was conducted. The updates to the system included modifications of services/maintenance and establishing modular permission approval in order to improve security. There were also minor tweaks aimed towards improving the user experience. Resources used: updated system prototype, adviser comments. Persons involved: all team members and thesis adviser.

April 12: The sixth official consultation was conducted. There were some needed updates in the system including interactions within the system, generation of audit logs and reports, as well as clarifying some of the intricacies of payment. Resources used: updated system prototype, adviser comments. Persons involved: all team members and thesis adviser.

Gantt Chart of Activities



Yellow indicate start and Blue indicate End. The summary of our activities. Listed are the activities and opposite them are their duration or periods of execution.

Resources

- **Hardware**

The following hardware resources were utilized in the development and implementation of the RRC Management System:

1. Laptops/Desktops – Used for system development, coding, and testing.

Personal Computer

- Device Name: Nitro 5 AN515-55

- Processor: Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz (8 CPUs), ~2.5GHz

- Graphics: NVIDIA® GeForce GTX 1650 Ti

- RAM: 16GB

- Display Resolution: 1920x1080

- Refresh Rate: 144Hz

- Storage Internal: 256GB SSD

- Storage External: 1TB HDD

2. Networking Equipment (Router, Ethernet Cables) – Ensures stable internet connectivity for system access and development.

- **Software**

The following software resources were essential for building and deploying the system:

1. Operating System (Windows) – Primary OS used for system development and testing.
2. ASP.NET – Framework for API development and business logic implementation.
3. Microsoft SQL Server (MS SQL) – Database management system for storing and retrieving data.
4. Visual Studio 2022– Integrated development environment (IDE) for coding and debugging.
5. Git/GitHub – Version control system for tracking project changes and collaboration.
6. Cloud Hosting Services (e.g., Azure, AWS) – Used for system deployment and ensuring accessibility.
7. Security Tools (AES-256 Encryption, Argon2 Password Hashing, Two-Factor Authentication) – Implemented for data protection and secure access control.
8. PayPal API – Integrated for enabling secure and seamless online payment processing for clients.

These resources were necessary to ensure the efficient development, testing, and deployment of the RRC Management System while maintaining scalability and security.